



Hybrid Imaging: Calcium Score and Myocardial Perfusion Imaging

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Coronary heart disease (CHD) remains the top cause of death due to cardiovascular conditions worldwide, with someone suffering a myocardial infarction every 40 seconds. This highlights the importance of non-invasive imaging technologies like myocardial perfusion imaging (MPI), which are crucial for detecting coronary artery disease (CAD) early, even before symptoms appear. However, the reliance solely on MPI has shifted due to its limitations in definitively ruling out atherosclerosis, leading to the adoption of hybrid imaging techniques. Hybrid imaging combines computed tomography (CT) with MPI techniques such as positron emission tomography (PET) and single photon emission computed tomography (SPECT). This integration, often within a single gantry system, enhances the diagnostic accuracy by allowing for attenuation correction (AC), acquisition of the coronary artery calcium score (CACS), and more precise tracing of radiotracer uptake. The built-in CT in modern MPI systems assists in these functions, which is essential for better diagnosis and risk assessment in patients. The addition of CACS to MPI, a method involving the assessment of calcified plaque in coronary arteries, notably enhances diagnostic and prognostic capabilities. CACS helps in identifying atherosclerosis and predicting potential cardiac events, facilitating personalized risk management and the initiation of tailored interventions like statins and aspirin. Such comprehensive imaging strategies not only improve the accuracy of detecting CAD but also help in stratifying patient risk more effectively. In this paper, we discuss how the incorporation of CAC into MPI protocols enhances the diagnostic sensitivity for detecting obstructive CAD, as evidenced by several studies where the addition of CAC to MPI has led to improved outcomes in diagnosing CAD. Moreover, CAC has been shown to unmask silent coronary atherosclerosis in patients with normal MPI results, highlighting its incremental diagnostic value. We will discuss the evolving role of hybrid imaging in guiding therapeutic decisions, particularly the use of statins for cardiovascular prevention. The integration of CAC assessment with MPI not only aids in the early detection and management of CAD but also optimizes therapeutic strategies, enhancing patient care through a more accurate and personalized approach. Such advancements underscore the need for further research to fully establish the benefits of combining CAC with MPI in the clinical assessment of cardiovascular risk.

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Introduction

Coronary heart disease stands as the leading cause of mortality attributable to cardiovascular disease in the

United States and globally, with one person experiencing a myocardial infarction (MI) every 40 seconds.¹ This underscores the critical role of non-invasive imaging in detecting coronary artery disease (CAD), even before the onset of clinical symptoms.^{2,3} Numerous studies have demonstrated that normal myocardial perfusion imaging (MPI) does not always exclude CAD, prompting a shift towards hybrid imaging and away from sole reliance on single imaging modalities.⁴⁻⁷

Hybrid imaging refers to imaging approaches that combine computed tomography (CT) with MPI as positron emission

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tomography (PET) and single photon emission computed tomography (SPECT).^{8,9} This integration can be accomplished either by using separate scanners for CT and radionuclide imaging or by incorporating both modalities into one gantry system.⁹ Nowadays, most of the newer MPI systems have a built-in CT scanner which can be used for several purposes.¹⁰ These encompass the attenuation correction (AC), the acquisition of the coronary artery calcium score (CACS), and the precise tracing of radiotracer uptake on PET and SPECT including fluorodeoxyglucose (FDG) hotspot imaging.^{10,11}

The utilization of CT-based attenuation correction (CTAC) improves the diagnostic accuracy of MPI, reduces artifacts, and enhances the specificity of PET and SPECT reconstructed images.¹²⁻¹⁶ Furthermore, incorporating CACS into MPI not only enhances risk stratification in SPECT¹⁷ but improves the overall diagnostic and prognostic value of MPI.^{7,18-20} This is attributed to the crucial role of CACS in identifying atherosclerosis²¹ and forecasting future cardiovascular events.²² As a result, this integration enables personalized risk factor modification for each patient including the initiation of tailored interventions such as statin and aspirin.²³⁻²⁶

In this review, we aim to explain the assessment of the CACS in hybrid imaging along with exploring its incremental diagnostic and prognostic value in CAD evaluation and discussing its impact on statin administration.

Calcium Score Assessment

Agaston Score

CACS has emerged as a powerful tool in cardiovascular preventive medicine, offering valuable insights into an individual's risk of future cardiac events and the presence of obstructive CAD.^{27,28} It offers a low-radiation, cost-effective way to accurately assess heart disease risk, particularly in asymptomatic individuals at intermediate risk.²⁹⁻³¹ Its high reliability, beyond traditional risk factors, helps tailor preventive measures and potentially decrease unnecessary interventions.²⁸

Several methods are available for the assessment of CACS: the Agatston Score (AS),³² calcium volume,³³ and relative calcium mass score.³⁴ The AS is the most popular and the most widely method used in databases and clinical practice. In brief, the relative calcium mass score entails multiplying the average attenuation of calcified plaque by the volume of each plaque in every image, thereby addressing the impact of partial volume. The calcium volume score is calculated by multiplying the number of voxels containing calcification by the volume of each voxel, making this method sensitive to partial volume. Notably, both the relative calcium mass and calcium volume scores have demonstrated better reproducibility than AS.^{33,35}

Visual Assessment From CT Attenuation Correction

The AS offers a well-established, semi-quantitative approach to estimating CAC, a key indicator of atherosclerotic burden.

It is obtained through breath-holding non-contrast gated cardiac CT with a tube voltage of 120 KVP.³⁶ This scoring system assigns weighted scores to individual calcified plaques, considering their area and density measured in Hounsfield Units (HU) on CT scans. Any pixel exceeding a threshold of 130 HU and having an area $\geq 1\text{mm}^2$ signifies calcification. The score for each lesion is computed by multiplying its automatically measured area by a density weighting factor (DWF), which adjusts for peak plaque attenuation within the lesion. Notably, the total AS is the sum of all individual lesion scores, irrespective of their location or distribution within the coronary arteries. This single numerical value reflects the overall extent of atherosclerosis. Importantly, calcium scores are influenced by age, gender, and race, requiring normalization for accurate interpretation.³⁷

Hybrid imaging cardiac nuclear/CT systems, in addition to adding prognostic value by performing AC, allow for both qualitative and quantitative assessment of a CACS.^{38,39} In contrast to traditional AS, it is acquired with a lower radiation dose (low dose length product) in a non-gated cardiac CT with free breathing or at end-expiration showcasing its efficiency in diagnostic procedures.³⁸ A study led by Mylonas et al.⁴⁰ explored the potential of using cardiac PET's AC to measure CACS. By setting appropriate HU thresholds for detecting calcium (including lower thresholds starting from 50), CTAC measurements could accurately quantify CAC. Importantly, the CTAC scores demonstrated a strong correlation with traditional AS, particularly for lower CAC scores. In a multicenter study of 492 patients, the feasibility and accuracy of visually estimating CAC using low-dose, non-gated CT have been evaluated. There was a significant correlation with AS, the gold standard for CAC quantification. Moreover, visually estimating CAC demonstrated excellent interobserver reproducibility, with nearly two-thirds of cases receiving identical scores and over 90% falling within one category of each other.⁴¹ Finally, CTAC allows interpreter to assess patient for extracardiac incidental findings and refer them for further investigations (Fig. 1).

Incremental Diagnostic Value of CAC Over MPI

Several recent studies have focused on assessing whether calcium score could add incremental diagnostic value to myocardial perfusion imaging. In one such study by Uretsky et al.,⁴² the addition of CAC results diminished the needs to perform rest SPECT-MPI studies among patients with probably normal or borderline stress-only SPECT MPI studies. Additionally, Schepis et al.⁴³ obtained CACS and SPECT-MPI on 77 patients with intermediate risk who were referred for coronary angiography. SPECT alone had a sensitivity and a specificity for the detection of significant CAD of 76% and 91%, respectively. However, combining it with CACS improved the sensitivity to 86%, with no significant increase in specificity. A similar study including 150 patients found that the addition of CAC to SPECT-MPI improved diagnostic

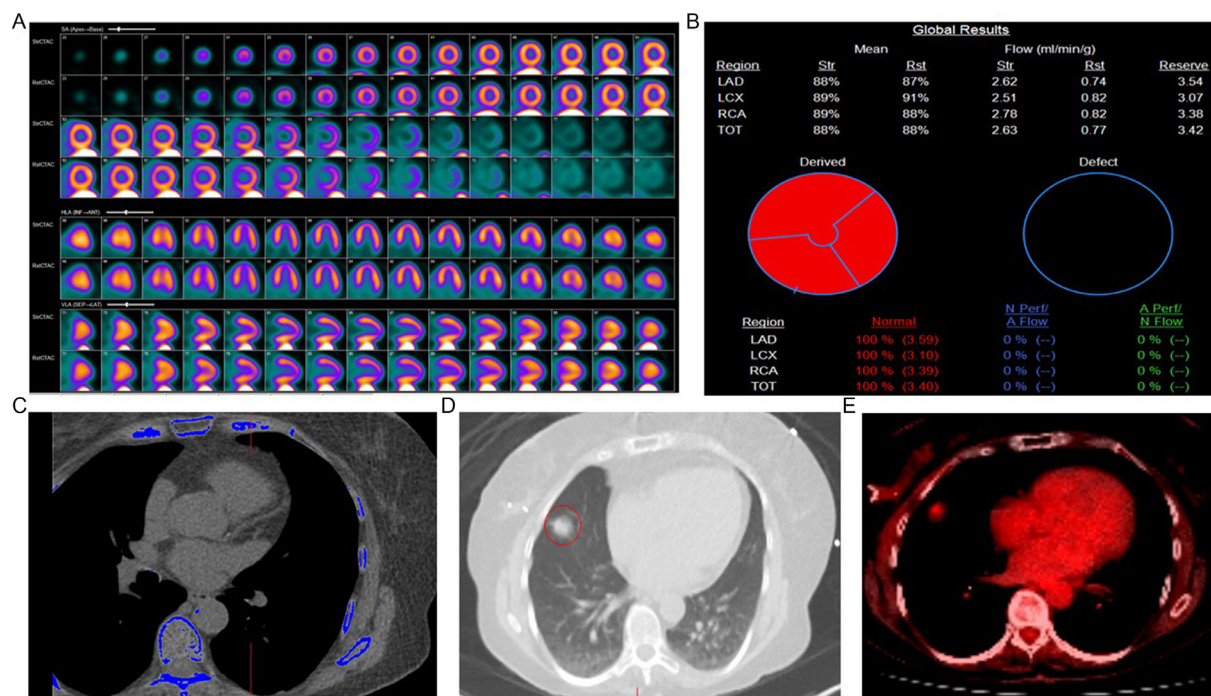


Figure 1 Seventy-one-year-old female with history of hypertension and dyslipidemia, presents to office for shortness of breath. She is unable to exercise. Patient is referred for hybrid cardiac PET and calcium score imaging. Results show normal perfusion (A), with normal myocardial flow reserve (MFR) in all territories (B). Calcium score is zero (C). However, upon reviewing the attenuation correction scan, a 1.2 cm non-calcified nodule can be seen in right lung (d). Based on the Fleischner criteria, the patient was referred to FDG PET, which showed mild FDG uptake in right middle lobe with no other FDG uptake (e). She underwent lung biopsy, which revealed a diagnosis of lung adenocarcinoma.

accuracy to detect obstructive CAD (AUC increase from 0.87 to 0.91).⁴⁴ Calcium score measurement can also help unmask the presence of silent coronary atherosclerotic disease. In fact, two studies found severe calcified coronary atherosclerosis (CACS>400) in 6.5% to 20% of patients with normal stress MPI.^{45,46} Similarly, a more recent study including around 5000 symptomatic patients with suspected CAD and normal SPECT-MPI found that 69% of patients had abnormal calcium score (CACS>0), while 15% had severely elevated calcium score (CACS>400)⁶ (Figure 2).

Prognostic Value of CAC In Patients Undergoing SPECT

Several studies have assessed the prognostic value of coronary artery calcium (CAC) scores in patients undergoing single-photon emission tomography imaging (SPECT). Chang and colleagues,¹⁸ performed a cohort study of 1,126 individuals who were asymptomatic or who had atypical chest pain with a median 6.9 years of follow-up. Two key findings were shown. First, ischemia as assessed by SPECT and atherosclerosis as assessed by CAC do not always overlap, as only 31% of patients with a CAC >400 had an abnormal SPECT scan. Second, CAC significantly improved the prognostic discrimination of SPECT. Among patients with a normal SPECT, the annual rate of cardiac events among those with a CAC score

of 0-10 was 0.7% versus 2.97% among those with a CAC >400. Uebli and colleagues⁴⁷ performed a cohort study of 260 patients with stable coronary artery disease (CAD) followed for a median of 5.4 years. In a multivariable model adjusted for age, sex, and other risk factors, both a summed rest score >2 and a CAC >400 were independently associated with a significantly increased risk of cardiac events (hazard ratios [HR] of 2.6 and 2.8 respectively).

Subsequently Ghadri et al.⁴⁸ published an analysis of 462 patients with known or suspected CAD and a median follow-up duration of 2.9 years. First, they provided additional evidence that ischemia weakly correlates with atherosclerosis, as most patients (59.3%) had discordant findings. The presence of matched findings (vessel calcification with a corresponding perfusion defect) was associated with a 3-fold elevation in the risk of cardiovascular events (HR of 3.2). These findings were further reinforced by Possner et al.⁴⁹ in 2015 in a cohort study of 275 patients with known or suspected CAD and a median follow-up duration of 2.9 years. In multivariable models adjusted for age, sex, and other risk factors, both perfusion defects identified by SPECT and CAC were independently associated with an increased risk of events.

Nappi and colleagues studied a cohort of 164 individuals with a low-to-medium pre-test likelihood of CAD followed up for a median of 6.8 years.⁵⁰ Both CAC score (assessed on a logarithmic scale) and SPECT-related ischemia (assessed via the summed difference score) were independently associated with cardiac events in multivariable Cox regression

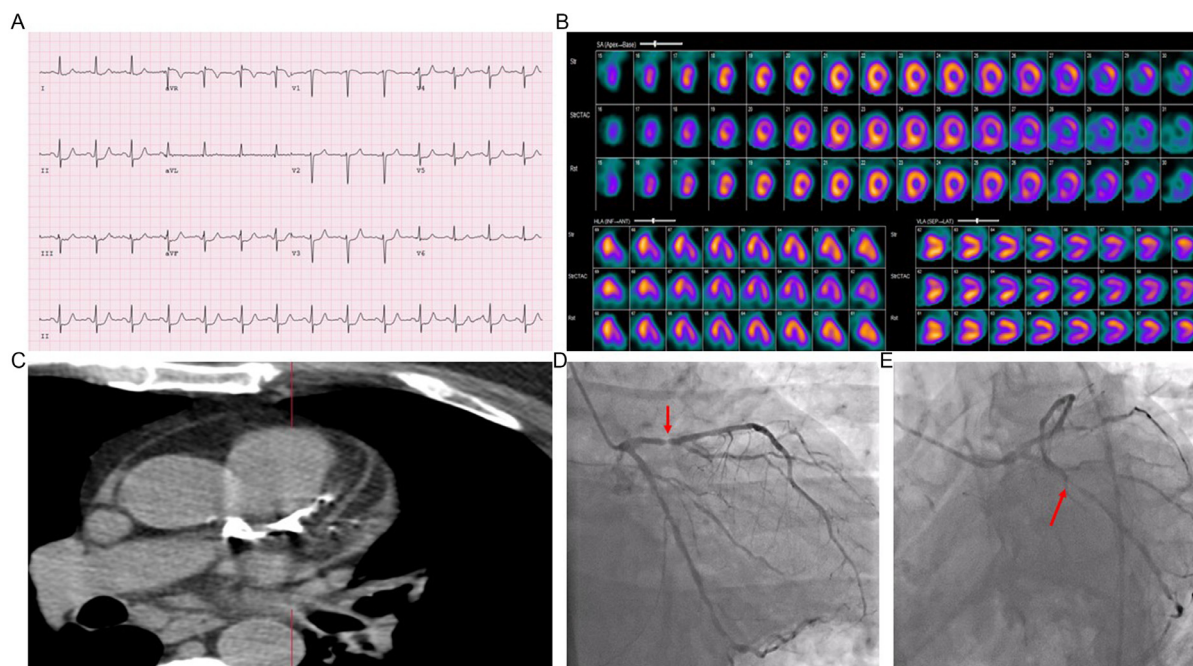


Figure 2 A 66-year-old man with history of hypertension, dyslipidemia, diabetes and peripheral artery disease, presented to the clinic with chest pain. Initial EKG and serial troponins were negative. Patient was referred for hybrid SPECT perfusion imaging and calcium score. His stress EKG showed mild ST depression in inferolateral leads (II-III-aVF) as well as poor R wave progression (A). On perfusion imaging, there was a mild apical-lateral perfusion defect during stress which normalized with rest imaging (B), as well as no wall motion abnormality and normal stress and rest EF. Calcium score testing revealed a CACS=2,827 with calcium in all three coronary vessels (C). Given EKG changes and elevated calcium score, patient was referred for invasive coronary angiography. Results show 80% stenosis in proximal LAD (D), as well as 90% stenosis in proximal LCx (E) and 60% stenosis in proximal RCA. Patient was referred for percutaneous coronary intervention of the LAD and LCx, after which symptoms significantly improved.

models. In addition, the authors also applied a decision-analytic rule to quantify the added clinical benefit of incorporating MPI measurements in patients with a CAC >0. They found that, over a range of decision thresholds ranging from 25% to 50%, adding MPI resulted in meaningful improvements in clinical management.

There is growing evidence to support the use of CAC scores obtained from attenuation correction images, as opposed to CAC scores obtained from a dedicated ECG-gated CT scan. Assessing the additive prognostic value of CAC in this setting is important as sometimes a CT attenuation correction (CTAC) may be all that is available. To address this question, Patchett et al. performed a cohort study of 1,047 individuals undergoing SPECT with 2.3 years of follow-up in whom only CTAC was available.⁵¹ Visible CAC observed on CTAC was independently associated with an increased risk of death or myocardial infarction (HR of 19.1) independently of ischemia seen on SPECT. This indicates that even when ECG-gated CAC scans are not available, visual CAC observed on CTAC images carries prognostic value that should be utilized.

Additionally, Engbers et al.⁶ analyzed a cohort of 4,897 asymptomatic patients without a history of CAD and found that both SPECT and CAC scores carried independent prognostic value. In multivariable models adjusted for age, sex,

and other risk factors, there was a stepwise increase in risk associated with small, moderate, and large perfusion defects on SPECT compared to no defect (HRs of 1.64, 2.21, and 3.74 respectively). Additionally, CAC scores of 1-99, 100-399, 400-999, and >1000 demonstrated stepwise increases in risk compared to a CAC of 0 (HRs of 1.35, 3.31, 4.87, and 7.57 respectively). In a subgroup analysis from the same cohort focused on those with a normal SPECT scan, similar findings of a stepwise increase in risk observed with CAC were seen.⁵² In 2022, Jayadeva and colleagues⁵³ documented the value of CAC scores in changing clinical management. They found that, after disclosing CAC results to treating physicians, a large number of patients experienced changes in their treatment regimens. This was particularly evident in those who had a negative MPI but a positive CAC, in whom treatment was frequently intensified. Hijazi et al.⁵⁴ also found in an analysis of 9,908 patients that both ischemia detected on SPECT and CAC were independently associated with increases in the prescription of aspirin and statins.

Other imaging biomarkers obtained from CT aside from CAC may also be valuable. Yang et al.⁵⁵ analyzed the additive prognostic value of epicardial fat volume when added to models adjusted for traditional risk factors, CAC, and MPI. They found that a high EFV (>108.3cm³) was independently

associated with a 3-fold elevation in the risk of adverse cardiovascular events (HR of 3.3).

Kammerman et al.⁵⁶ aimed to assess the “warranty period” (annual MACE rate <1%, including late revascularization, non-fatal MI and cardiac death offered by a normal SPECT scan and whether it could be extended by additionally incorporating CAC. They found that a normal SPECT coupled with a CAC score of <400 was associated with a 10-year warranty period, a normal SPECT coupled with a CAC score of 400-999 was associated with a 4-year warranty period, and a normal SPECT but a CAC of $\geq 1,000$ was associated with such a high rate of events at 1 year (5.9%) so that no warranty period could be offered. Feher et al. also compared machine-learning (ML) models with and without CAC.⁵⁷ They found that ML incorporating CAC led to a statistically significant improvement in the area under the curve (AUC) of about 0.02 (0.87 for ML with CAC versus 0.85 for ML without CAC).

In addition, deep learning can measure CACS from CTAC. Feher et al.⁵⁸ investigated the clinical utility of CAC obtained from CTAC and found that it was associated with significant increases in adverse cardiovascular events. Furthermore, they also found that the AUC for visually estimated CAC was like the AUC for quantitative CAC (0.694 vs 0.691 respectively).

Prognostic Value of CAC in Patients Undergoing PET

As with SPECT, a number of studies have assessed the added prognostic value of CAC to positron emission tomography (PET) imaging. Schenker et al.²⁰ in 2008 studied a cohort of 695 patients at intermediate risk of CAD followed-up for a median of 1.5 years, there were two notable findings. First, as with SPECT, large numbers of patients with a normal MPI had a CAC > 0. For example, among patients with a CAC $\geq 1,000$, nearly half (49.4%) had a non-ischemic PET scan (where ischemia was defined by an SDS ≥ 2). Conversely, among patients with a CAC of 0, nearly 16% had ischemia detected by PET. Second, in Cox regression models, both an SDS ≥ 2 and a CAC $\geq 1,000$ independently predicted an increased risk of cardiovascular events (HR of 2.34 and 1.88 respectively).

Mylonas et al.⁴⁰ assessed the correlation between CAC assessed on CTAC versus ECG-gated CCTA and found a high correlation coefficient of 0.954, indicating that CTAC-assessed CAC is generally quite similar to CAC assessed via dedicated ECG-gated CCTA. However, it is important to note that, although differences were small at the lower end of the CAC spectrum, they became larger at higher values of CAC. In sum, this indicates that a low CAC score obtained from CTAC is likely quite reliable, but that a high CAC score may need further verification.

Naya and colleagues assessed the prognostic value of CAC compared to coronary flow reserve (MFR) in 901 patients with a normal MPI.¹⁹ They found that, compared to a multi-variable clinical model incorporating the Duke clinical risk

score, the addition of either CAC or MFR improved diagnostic discrimination as assessed by the likelihood ratio test comparing the 2 models. However, although the addition of MFR to CAC improved discrimination, the converse (addition of CAC to MFR) did not. Nevertheless, the study had a relatively small number of events (57) and may have been underpowered.

Le et al.⁵⁹ of 5,528 patients without CAD referred for PET evaluation. The absence of CAC (on CTAC) was associated with a 55% lower rate of adverse events (HR of 0.45), an association which was statistically significant even after adding ischemic burden to the model. Subsequently, Dekker et al. also showed that CAC as quantified by a deep-learning model from CTAC obtained during PET MPI was independently associated with cardiovascular adverse events.⁶⁰

Aljizeeri and colleagues⁶¹ assessed the combined value of myocardial flow reserve (MFR) and a CAC of 0. They found that in patients with a CAC of 0, MFR provided further risk stratification as those with a MFR above or below 2 had event rates of 0.65% versus 0.17% respectively. This refinement in risk stratification was also evident in other CAC categories, including those with a CAC of ≥ 400 . Similarly, the risk of events in those with an MFR <2 differed significantly according to CAC (from 0.65% in those with a CAC of 0 to 5.71% in those with a CAC of ≥ 400). However, compared to 2 multivariable models that incorporated CAC and MFR separately, a model containing both parameters did not significantly improve risk discrimination (AUC of 0.833 and 0.828 versus 0.844 respectively). In a subsequent study of 2,507 patients followed for a median of 3.9 years,⁶² Miller et al. showed that CAC, MPI-ischemia (as assessed by an automatically-quantified total perfusion defect), and MFR were associated the risk of cardiovascular events. Compared to previous studies, this latter study by Miller et al. had a greater number of events (594) and therefore likely had greater statistical power to detect the independent prognostic value of each parameter.

Table summarizes all studies assessing the added prognostic value of CT parameters (mainly CACS and visual calcium score) to SPECT and PET MPI.

Can Hybrid Imaging Guide Pharmacotherapy?

The role of statins in cardiovascular primary and secondary prevention is well established and endorsed by most medical societies,^{63,64} and there is growing evidence supporting the role of CAC measurement in identifying patients in need of therapy. One such study looked at 950 patients who met criteria for the JUPITER (Justification for the Use of Statins in Prevention: an Intervention Trial Evaluating Rosuvastatin) clinical trial. Investigators found a clinically significant difference in the 5-year Number Needed to Treat to prevent one cardiovascular event between individuals with CAC=0 and CAC>100 (124 and 19, respectively).⁶⁵ Additionally, in the context of the 2013 ACC/AHA, Nasir et al. conducted an

Table Summary of Studies Assessing the Added Prognostic Value of CT Parameters to SPECT and PET MPI

Study	Sample Size	Follow-up Duration (years)	Study Question	Finding
SPECT				
Chang et al., 2009 ¹⁸	1,126	6.9	Does integrating CAC with SPECT MPI improve prognostic accuracy?	CAC and SPECT MPI are independently associated with cardiovascular risk and, when combined, provide improved prognostication.
Uebleis et al., 2009 ⁴⁷	260	5.4	In patients with stable CAD, does adding CAC improve prognostic value over SPECT MPI?	In patients with stable CAD, CAC improved prognostication when added to SPECT MPI.
Ghadri et al., 2013 ⁴⁸	462	2.9	Does the combination of a SPECT perfusion defect and co-localized vessel calcification portend greater risk than either finding alone?	Patients with a calcified vessel and a corresponding perfusion defect in the same territory experience a greater rate of events compared to patients with either finding.
Possner et al., 2015 ⁴⁹	275	2.9	Does the addition of EFV and IFV to CAC and MPI improve risk discrimination?	EFV and IFV did not result in statistically significant improvements in risk discrimination after incorporating MPI and CAC.
Nappi et al., 2016 ⁵⁰	164	6.8	In patients with suspected CAD, what is the long-term prognostic value of stress SPECT, CAC, and CCTA?	Both stress MPI and CAC findings were independently associated with cardiovascular events. The addition of CCTA-assessed stenosis was not statistically significant.
Engbers et al., 2016 ⁵²	3705	2.8	In patients with a normal MPI, does CAC add prognostic value?	CAC had a stepwise association with the risk of cardiovascular events in patients with a normal MPI
Engbers et al., 2016 (B) ⁶	4897	2.6	In patients with suspected CAD, what is the added prognostic value of CAC over SPECT MPI?	CAC and ischemia on SPECT, independent of traditional risk factors and of one another, had a stepwise association with the risk of cardiovascular events
Patchet et al., 2016 ⁵¹	1047	2.3	Does CAC as visualized on CTAC carry prognostic value?	The presence of CAC on CTAC was associated with an increased risk of cardiovascular events
Jayadeva et al., 2022 ⁵³	281	N/A	Does knowledge of CAC after MPI change clinical management?	Knowledge of CAC scores was followed by changes in clinical management for a number of patients. This was particularly evident in those with a negative MPI and positive CAC, in whom preventive treatment was intensified.
Hijazi et al., 2023 ⁵⁴	9908	N/A	Do CAC scores and MPI scans associate with changes in management?	The presence of CAC and an abnormal MPI were associated with increased prescription of statins and aspirin.
Yang et al., 2023 ⁵⁵	290	Unreported	Does EFV predict an increased risk of cardiac events?	Independent of CAC, MPI, and traditional risk factors, EFV was associated with an increased risk of cardiac events
Kamerman et al., 2023 ⁵⁶	1375	10.3	Does a CAC of zero in addition to a normal SPECT scan extend the warranty period associated with the scan?	In patients with a normal SPECT, CAC helped further refine the warranty period associated with the scan, such that a CAC of 0 resulted in a 10-year warranty period and a CAC of 1000 or more meant that no warranty period could be offered
Feher et al., 2023 ⁵⁷	4770	2	Does the combination of CAC in a machine learning algorithm improve prognostic value over machine learning without CAC or CAC without machine learning?	The combination of CAC and machine learning provided better risk prediction than using either separately.

Table (Continued)

Study	Sample Size	Follow-up Duration (years)	Study Question	Finding
Feher et al., 2024 ⁵⁸	4236	2.1	How does use of visually estimated CAC from CTAC compare to quantitative CAC?	Use of visually estimated CAC and quantified CAC resulted in a similar ROC AUC.
PET Schenker et al., 2008 ²⁰	695	1.5	Does adding CAC to PET MPI improve prognostic value?	In patients with or without ischemia on PET, increased CAC is associated with an increased risk of cardiovascular events.
Mylonas et al., 2012 ⁴⁰	91	N/A	How closely does CTAC-assessed CAC correlate with CCTA-derived CAC?	CAC assessed via CTAC images was highly correlated with CAC assessed via CCTA, but the correlation weakened at higher values of CAC
Naya et al., 2013 ¹⁹	901	1.5	In patients with a normal MPI, what is the contribution of CFR, CAC, and their combination to assessing prognosis?	In patients with a normal MPI, CFR improved risk discrimination to a greater extent than CAC. The combination of the two parameters was not associated with a statistically significant improvement in risk discrimination.
Le et al., 2020 ⁵⁹	5528	1.0	In patients undergoing PET, does CAC from CTAC add prognostic value?	Patients with no CAC on CTAC were at a very low risk of revascularization or cardiovascular events
Dekker et al., 2021 ⁶⁰	747	3.6	Does CAC from CTAC as quantified by a deep-learning algorithm add prognostic value to PET MPI?	CAC scores obtained from a deep-learning model were significantly associated with the risk of cardiovascular events
Aljizeeri et al., 2021 ⁶¹	4008	1.9	What is the prognostic value of CFR and CAC separately and in combination?	Both CFR and CAC were predictors of long-term prognosis, but the incorporation of both did not result in statistically significant improvements in predictive value.
Miller et al., 2022 ⁶²	2507	3.9	Do PET-detected ischemia, CFR, and CAC carry prognostic value independent of one another?	All 3 parameters carried prognostic value both alone and in combination with one another.

CT: Computed tomography; SPECT: Single photon emission tomography; PET: Positron emission tomography; MPI: Myocardial perfusion imaging; CAC: Coronary artery calcium; EFV: Epicardial fat volume; CTAC: Computed tomography attenuation correction; CFR: Coronary flow reserve; ROC: Receiver operating characteristics; AUC: Area under the curve; EFV: Epicardial fat volume; CTAC: Computed tomography attenuation correction; CFR: Coronary flow reserve; ROC: Receiver operating characteristics; AUC: Area under the Curve; N/A: Not applicable

analysis that included MESA participants with a 10-year ASCVD risk between 5% and 7.5% using the Pooled Cohort Equation. Results showed that CAC = 0 was associated with an observed ASCVD event rates below the guideline-based threshold of statin benefit (~4.5%). Conversely, any CAC > 0 was associated with events consistent with net benefit from statin therapy (~10.5%).^{66,67} A similar analysis was conducted with regard to the indication to start statin therapy and found that CAC added to risk stratification of both statin-recommended and non-statin recommended groups.⁶⁸ A more recent study by Patel et al.⁶⁹ found that among a cohort of 1688 MESA participants with intermediate ASCVD found similar results. These studies highlighted the important role of calcium score testing in intermediate risk population with regards to statin initiation. This is evidenced by the recommendation in the 2018 AHA/ACC guidelines for blood cholesterol management which recommend the use of calcium score testing in intermediate-risk adults where there is uncertainty regarding initiation of statin therapy (IIa B-NR).⁶³ As more interest is directed towards dual MPI and calcium score imaging, few studies have investigated the potential benefit of the combination of these two modalities with regards to statin therapy. One such study by To et al.⁷⁰ found that among 311 patients with normal SPECT-MPI and CAC > 300, initiation of post-imaging moderate-to-high dose statin was an independent negative predictor for 24-month MACE.

Conclusion

By providing both anatomical as well as functional assessment, Hybrid imaging with calcium score and myocardial perfusion imaging seems to be a promising tool for diagnosis and risk stratification of patients with suspected or known coronary artery disease. Perhaps more trials are needed assessing the incremental diagnostic value of CAC on top of MPI, especially in patients with no ischemia on SPECT or PET.

Declaration of competing interest

Dr AL-Mallah is a consultant for Jubilant and Pfizer. He received a research grant from Siemens. All other authors have no disclosures.

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References

1. Tsao CW, Aday AW, Almarzooq ZI, et al: Heart Disease and Stroke Statistics-2023 Update: A Report From the American Heart Association. *Circulation* 147:e93-e21, 2023
2. Knuuti J, Wijns W, Saraste A, et al: 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J* 41:407-477, 2020
3. Gulati M, Levy PD, Mukherjee D, et al: 2021 AHA/ACC/AASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Cardiovasc Comput Tomogr* 16:54-122, 2022
4. Patel KK, Peri-Okonny PA, Qarajeh R, et al: Prognostic Relationship Between Coronary Artery Calcium Score, Perfusion Defects, and Myocardial Blood Flow Reserve in Patients With Suspected Coronary Artery Disease. *Circ Cardiovasc Imaging*. 15:e012599, 2022
5. Kalia NK, Miller LG, Nasir K, Blumenthal RS, Agrawal N, Budoff MJ: Visualizing coronary calcium is associated with improvements in adherence to statin therapy. *Atherosclerosis* 185:394-399, 2006
6. Engbers EM, Timmer JR, Ottervanger JP, Mouden M, Knollemans S, Jager PL: Prognostic Value of Coronary Artery Calcium Scoring in Addition to Single-Photon Emission Computed Tomographic Myocardial Perfusion Imaging in Symptomatic Patients. *Circ Cardiovasc Imaging*. 9. <https://doi.org/10.1161/circimaging.115.003966>, 2016
7. He ZX, Hedrick TD, Pratt CM, et al: Severity of coronary artery calcification by electron beam computed tomography predicts silent myocardial ischemia. *Circulation* 101:244-251, 2000
8. Gaemperli O, Bengel FM, Kaufmann PA: Cardiac hybrid imaging. *Eur Heart J* 32:2100-2108, 2011
9. Dorbala S, Di Carli MF, Delbeke D, et al: SNMMI/ASNC/SCCT guideline for cardiac SPECT/CT and PET/CT 1.0. *J Nucl Med* 54:1485-1507, 2013
10. Al-Mallah MH, Bateman TM, Branch KR, et al: 2022 ASNC/AAPM/SCCT/SNMMI guideline for the use of CT in hybrid nuclear/CT cardiac imaging. *J Nucl Cardiol* 29:3491-3535, 2022
11. Chen W, Kim J, Molchanova-Cook OP, Dilsizian V: The potential of FDG PET/CT for early diagnosis of cardiac device and prosthetic valve infection before morphologic damages ensue. *Curr Cardiol Rep* 16:459, 2014
12. Genovesi D, Giorgetti A, Gimelli A, et al: Impact of attenuation correction and gated acquisition in SPECT myocardial perfusion imaging: results of the multicentre SPAG (SPECT Attenuation Correction vs Gated) study. *Eur J Nucl Med Mol Imaging* 38:1890-1898, 2011
13. Huang JY, Huang CK, Yen RF, et al: Diagnostic Performance of Attenuation-Corrected Myocardial Perfusion Imaging for Coronary Artery Disease: A Systematic Review and Meta-Analysis. *J Nucl Med* 57:1893-1898, 2016
14. Masood Y, Liu YH, Depuey G, et al: Clinical validation of SPECT attenuation correction using x-ray computed tomography-derived attenuation maps: multicenter clinical trial with angiographic correlation. *J Nucl Cardiol* 12:676-686, 2005
15. Sharma P, Patel CD, Karunanihi S, Maharjan S, Malhotra A: Comparative accuracy of CT attenuation-corrected and non-attenuation-corrected SPECT myocardial perfusion imaging. *Clin Nucl Med* 37:332-338, 2012
16. Utsunomiya D, Tomiguchi S, Shiraishi S, et al: Initial experience with X-ray CT based attenuation correction in myocardial perfusion SPECT imaging using a combined SPECT/CT system. *Ann Nucl Med* 19:485-489, 2005
17. Trpkov C, Savtchenko A, Liang Z, et al: Visually estimated coronary artery calcium score improves SPECT-MPI risk stratification. *Int J Cardiol Heart Vasc* 35:100827, 2021

18. Chang SM, Nabi F, Xu J, et al: The coronary artery calcium score and stress myocardial perfusion imaging provide independent and complementary prediction of cardiac risk. *J Am Coll Cardiol* 54:1872-1882, 2009
19. Naya M, Murthy VL, Foster CR, et al: Prognostic interplay of coronary artery calcification and underlying vascular dysfunction in patients with suspected coronary artery disease. *J Am Coll Cardiol* 61:2098-2106, 2013
20. Schenker MP, Dorbala S, Hong EC, et al: Interrelation of coronary calcification, myocardial ischemia, and outcomes in patients with intermediate likelihood of coronary artery disease: a combined positron emission tomography/computed tomography study. *Circulation* 117:1693-1700, 2008
21. Rumberger JA, Simons DB, Fitzpatrick LA, Sheedy PF, Schwartz RS: Coronary artery calcium area by electron-beam computed tomography and coronary atherosclerotic plaque area. A histopathologic correlative study. *Circulation*. 92:2157-2162, 1995
22. Budoff MJ, Young R, Burke G, et al: Ten-year association of coronary artery calcium with atherosclerotic cardiovascular disease (ASCVD) events: the multi-ethnic study of atherosclerosis (MESA). *Eur Heart J*. 39:2401-2408, 2018
23. Golub I, Lakshmanan S, Dahal S, Budoff MJ: Utilizing coronary artery calcium to guide statin use. *Atherosclerosis* 326:17-24, 2021
24. Cainzos-Achirica M, Miedema MD, McEvoy JW, et al: Coronary Artery Calcium for Personalized Allocation of Aspirin in Primary Prevention of Cardiovascular Disease in 2019: The MESA Study (Multi-Ethnic Study of Atherosclerosis). *Circulation*. 141:1541-1553, 2020
25. Taylor AJ, Bindeman J, Feuerstein I, et al: Community-based provision of statin and aspirin after the detection of coronary artery calcium within a community-based screening cohort. *J Am Coll Cardiol* 51:1337-1341, 2008
26. Chaikriangkrai K, Palamaner Subash Shantha G, Jhun HY, et al: Prognostic Value of Coronary Artery Calcium Score in Acute Chest Pain Patients Without Known Coronary Artery Disease: Systematic Review and Meta-analysis. *Ann Emerg Med* 68:659-670, 2016
27. Greenland P, Blaha MJ, Budoff MJ, Erbel R, Watson KE: Coronary Calcium Score and Cardiovascular Risk. *Journal of the American College of Cardiology* 72:434-447, 2018
28. Members TF, Montalescot G, Sechtem U, et al: 2013 ESC guidelines on the management of stable coronary artery disease: The Task Force on the management of stable coronary artery disease of the European Society of Cardiology. *European Heart Journal* 34:2949-3003, 2013
29. Roberts ET, Horne A, Martin SS, et al: Cost-effectiveness of coronary artery calcium testing for coronary heart and cardiovascular disease risk prediction to guide statin allocation: the Multi-Ethnic Study of Atherosclerosis (MESA). *PLoS One* 10:e0116377, 2015
30. Baron KB, Choi AD, Chen MY: Low Radiation Dose Calcium Scoring: Evidence and Techniques. *Curr Cardiovasc Imaging Rep* 9:12, 2016
31. Nasir K, Clouse M: Role of Nonenhanced Multidetector CT Coronary Artery Calcium Testing in Asymptomatic and Symptomatic Individuals. *Radiology* 264:637-649, 2012
32. Agatston AS, Janowitz WR, Hildner FJ, Zusmer NR, Viamonte M, Detrano R: Quantification of coronary artery calcium using ultrafast computed tomography. *Journal of the American College of Cardiology* 15:827-832, 1990
33. McCollough CH, Ulzheimer S, Halliburton SS, Shanneik K, White RD, Kalender WA: Coronary Artery Calcium: A Multi-institutional, Multimanufacturer International Standard for Quantification at Cardiac CT. *Radiology* 243:527-538, 2007
34. Rumberger JA, Kaufman L: A rosetta stone for coronary calcium risk stratification: agatston, volume, and mass scores in 11,490 individuals. *AJR Am J Roentgenol* 181:743-748, 2003
35. Knudsen AD, Fuchs A, Kühl JT, et al: Coronary artery calcium assessed with calibrated mass scoring in asymptomatic individuals: results from the Copenhagen General Population Study. *European Radiology* 28:4607-4614, 2018
36. Budoff MJ, Achenbach S, Blumenthal RS, et al: Assessment of Coronary Artery Disease by Cardiac Computed Tomography. *Circulation* 114:1761-1791, 2006
37. Kondos GT, Hoff JA, Sevrakov A, et al: Electron-Beam Tomography Coronary Artery Calcium and Cardiac Events. *Circulation* 107:2571-2576, 2003
38. Al-Mallah MH, Bateman TM, Branch KR, et al: 2022 ASNC/AAPM/SCCT/SNMMI guideline for the use of CT in hybrid nuclear/CT cardiac imaging. *Journal of Nuclear Cardiology* 29:3491-3535, 2022
39. Qureshi WT, Alirhayim Z, Khalid F, MH Al-Mallah: Prognostic value of extracardiac incidental findings on attenuation correction cardiac computed tomography. *J Nucl Cardiol* 23:1266-1274, 2016
40. Mylonas I, Kazmi M, Fuller L, et al: Measuring coronary artery calcification using positron emission tomography-computed tomography attenuation correction images. *Eur Heart J Cardiovasc Imaging* 13:786-792, 2012
41. Einstein AJ, Johnson LL, Bokhari S, et al: Agreement of visual estimation of coronary artery calcium from low-dose CT attenuation correction scans in hybrid PET/CT and SPECT/CT with standard Agatston score. *J Am Coll Cardiol* 56:1914-1921, 2010
42. Uretsky S, Cohen R, Argulian E, et al: Combining stress-only myocardial perfusion imaging with coronary calcium scanning as a new paradigm for initial patient work-up: an exploratory analysis. *J Nucl Cardiol* 22:89-97, 2015
43. Schepis T, Gaemperli O, Koepfli P, et al: Added value of coronary artery calcium score as an adjunct to gated SPECT for the evaluation of coronary artery disease in an intermediate-risk population. *J Nucl Med* 48:1424-1430, 2007
44. Dekker M, Waissi F, Bank IEM, et al: Automated calcium scores collected during myocardial perfusion imaging improve identification of obstructive coronary artery disease. *Int J Cardiol Heart Vasc* 26:100434, 2020
45. Thompson RC, McGhie AI, Moser KW, et al: Clinical utility of coronary calcium scoring after nonischemic myocardial perfusion imaging. *J Nucl Cardiol* 12:392-400, 2005
46. Berman DS, Wong ND, Gransar H, et al: Relationship between stress-induced myocardial ischemia and atherosclerosis measured by coronary calcium tomography. *J Am Coll Cardiol* 44:923-930, 2004
47. Uebles C, Becker A, Griesshammer I, et al: Stable coronary artery disease: prognostic value of myocardial perfusion SPECT in relation to coronary calcium scoring—long-term follow-up. *Radiology* 252:682-690, 2009
48. Ghadri JR, Fiechter M, Fuchs TA, et al: Registry for the Evaluation of the Prognostic value of a novel integrated imaging approach combining Single Photon Emission Computed Tomography with coronary calcification imaging (REPROSPECT). *Eur Heart J Cardiovasc Imaging* 14:374-380, 2013
49. Possner M, Liga R, Gaisl T, et al: Quantification of epicardial and intrathoracic fat volume does not provide an added prognostic value as an adjunct to coronary artery calcium score and myocardial perfusion single-photon emission computed tomography. *Eur Heart J Cardiovasc Imaging* 17:885-891, 2016
50. Robinson AA, Bourque JM: Combined non-invasive imaging for predicting cardiovascular events : Is three a crowd? *J Nucl Cardiol* 25:842-844, 2018
51. Patchett ND, Pawar S, Miller EJ: Visual identification of coronary calcifications on attenuation correction CT improves diagnostic accuracy of SPECT/CT myocardial perfusion imaging. *J Nucl Cardiol* 24:711-720, 2017
52. Engbers EM, Timmer JR, Ottervanger JP, Mouden M, Knollemans S, Jager PL: Impact of Gender on the Prognostic Value of Coronary Artery Calcium in Symptomatic Patients With Normal Single-Photon Emission Computed Tomography Myocardial Perfusion. *Am J Cardiol* 118:1611-1615, 2016
53. Jayadeva PS, Stowers S, Tang EW, et al: The impact of coronary calcium score as an addition to myocardial perfusion imaging in altering clinical management (ICCAMP trial). *J Nucl Cardiol* 30:1004-1018, 2023
54. Hijazi W, Feng Y, Southern DA, et al: Impact of Myocardial Perfusion and Coronary Calcium on Medical Management for Coronary Artery Disease. *Eur Heart J Cardiovasc Imaging* 2023. <https://doi.org/10.1093/ehjci/ead288>
55. Yang L, Yu W, Wan P, et al: Epicardial fat volume, an independent risk factor for major adverse cardiovascular events, had an incremental

- prognostic value to myocardial perfusion imaging in Chinese populations with suspected or known coronary artery disease with a normal left ventricular ejection fraction. *Front Cardiovasc Med* 10:1261215, 2023
56. Kamerman M, van Dijk JD, Timmer JR, et al: The incremental value of coronary artery calcium score in predicting long-term prognosis and defining the warranty period of normal adenosine stress-only myocardial perfusion imaging using CZT SPECT. *J Nucl Cardiol* 30:2692-2701, 2023
 57. Feher A, Pieszko K, Miller R, et al: Integration of coronary artery calcium scoring from CT attenuation scans by machine learning improves prediction of adverse cardiovascular events in patients undergoing SPECT/CT myocardial perfusion imaging. *J Nucl Cardiol* 30:590-603, 2023
 58. Feher A, Pieszko K, Shanbhag A, et al: Comparison of the prognostic value between quantification and visual estimation of coronary calcification from attenuation CT in patients undergoing SPECT myocardial perfusion imaging. *Int J Cardiovasc Imaging* 40:185-193, 2024
 59. Le VT, Knight S, Min DB, et al: Absence of Coronary Artery Calcium During Positron Emission Tomography Stress Testing in Patients Without Known Coronary Artery Disease Identifies Individuals With Very Low Risk of Cardiac Events. *Circ Cardiovasc Imaging*. Jan 2020;13:e009907
 60. Dekker M, Waissi F, Bank IEM, et al: The prognostic value of automated coronary calcium derived by a deep learning approach on non-ECG gated CT images from (82)Rb-PET/CT myocardial perfusion imaging. *Int J Cardiol* 329:9-15, 2021
 61. Aljizeeri A, Ahmed AI, Alfari MA, et al: Myocardial Flow Reserve and Coronary Calcification in Prognosis of Patients With Suspected Coronary Artery Disease. *JACC Cardiovasc Imaging* 14:2443-2452, 2021
 62. Miller RJH, Han D, Singh A, et al: Relationship between ischaemia, coronary artery calcium scores, and major adverse cardiovascular events. *Eur Heart J Cardiovasc Imaging* 23:1423-1433, 2022
 63. Grundy SM, Stone NJ, Bailey AL, et al: 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation* 139:e1082-e1143, 2019
 64. Arnett DK, Blumenthal RS, Albert MA, et al: 2019 ACC/AHA Guideline on the Primary Prevention of Cardiovascular Disease: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation* 140:e596-e646, 2019
 65. Blaha MJ, Budoff MJ, DeFilippis AP, et al: Associations between C-reactive protein, coronary artery calcium, and cardiovascular events: implications for the JUPITER population from MESA, a population-based cohort study. *Lancet* 378:684-692, 2011
 66. Nasir K, Bittencourt MS, Blaha MJ, et al: Implications of Coronary Artery Calcium Testing Among Statin Candidates According to American College of Cardiology/American Heart Association Cholesterol Management Guidelines: MESA (Multi-Ethnic Study of Atherosclerosis). *J Am Coll Cardiol* 66:1657-1668, 2015
 67. Goff DC Jr., Lloyd-Jones DM, Bennett G, et al: 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol* 63:2935-2959, 2014
 68. Mahabadi AA, Möhlenkamp S, Lehmann N, et al: CAC Score Improves Coronary and CV Risk Assessment Above Statin Indication by ESC and AHA/ACC Primary Prevention Guidelines. *JACC: Cardiovascular Imaging* 10:143-153, 2017
 69. Patel J, Pallazola VA, Dudum R, et al: Assessment of Coronary Artery Calcium Scoring to Guide Statin Therapy Allocation According to Risk-Enhancing Factors: The Multi-Ethnic Study of Atherosclerosis. *JAMA Cardiology* 6:1161-1170, 2021
 70. To YH, Tan XM, Low CL, Min HH, Yew MS: Impact of statin use on major adverse cardiovascular events in patients with normal stress myocardial perfusion imaging and elevated coronary artery calcium score. *Singapore Med J* 64:109-114, 2023